Block-2–only weight decay reverses the story: protected parents, rising $R(t)$, and decay that again favors composition

exp. 6 — identical to exp. 4 except phase-2 λ acts on V_2 only — 8 α × 8 λ × 3 seeds

Design

Identical to exp. 4 (converged-$d2$ pretrain, $V_1 \to \alpha V_1$ at $d4$ onset setting $R_0 \in [0.04, 0.57]$, everything trainable, $d2$ losses kept, fresh optimizer, exact Walsh, per-cell measurements every 500 steps) with **one change**: the phase-2 weight decay is applied *only to the second-block weights* V_2; embedding, V_1 and readout are decay-free. The circuit under study is thus regularized, while the representation that feeds it is protected — the control that exp. 4 identified as missing. The 10k pretraining checkpoints are saved per seed (`accessibility_ckpts/`, also on Drive). Every cell solves $d4$ ($|\hat{f}_4| = 1.000$ throughout).

Findings

1. Decay again favors the compositional circuit, with the boundary at R_c. Right of R_c the parent share now *grows* with λ: at $R_0 = 0.52$, $f_{2+2} = 0.103$ (λ=0) $\to$ 0.175 (10^{-3}) $\to$ 0.230 (10^{-2}) $\to$ 0.385 ($3{\cdot}10^{-2}$); left of R_c it stays ≤ 0.01 at every λ (Figs. 1, 2 left). This is the frozen-case direction restored — global decay (exp. 4) drove the same cells to $f_{2+2} = 0$. The magnitude still trails the frozen benchmark (0.385 vs. 0.90 at matched R, λ): with the task solved within the first few hundred steps, the early singleton lock-in owns most of the output, and V_2-decay reworks only part of it.

2. The representation drift reverses. With V_1 protected, λ now pushes R_{final} *up*, not down (Fig. 2, right): from $R_0 = 0.04$, R_{final} rises 0.14 $\to$ 0.29 as λ: $0 \to 10^{-2}$; supercritical starts stay supercritical at every λ ($R_{\text{final}} \approx 0.41$–$0.46$ at $R_0 = 0.52$, vs. 0.21 under global decay). Mechanically: taxing block-2 circuits shifts gradient pressure into the untaxed V_1/embedding, strengthening the parents — the opposite of exp. 4's inverse-α erosion.

3. No dismantling transients. The phase plane (Fig. 3) shows trajectories moving *up* in f_{2+2} under decay (ends above starts), never the build-then-demolish loops of exp. 4: with $R(t)$ held supercritical, decay works *for* the compositional circuit throughout training.

4. Strong decay exits through a different door. At $\lambda = 10^{-1}$ (all cells) and a few at $3{\cdot}10^{-2}$, block 2 is suppressed entirely — no neuron contributes to $d4$ ($n_{\text{rel}} = 0$; NaN cells, dropped from the diagrams) — yet $\hat{f}_4 = 1.0$: the readout solves the task directly from the frozen stream's own $d4$ leak ($\|q_{1234}\| \approx 0.105\,\alpha$ from the block-1 corner units), the same bypass seen at $\alpha = 8$ in the frozen experiment. The "circuit choice" question dissolves there rather than being answered.

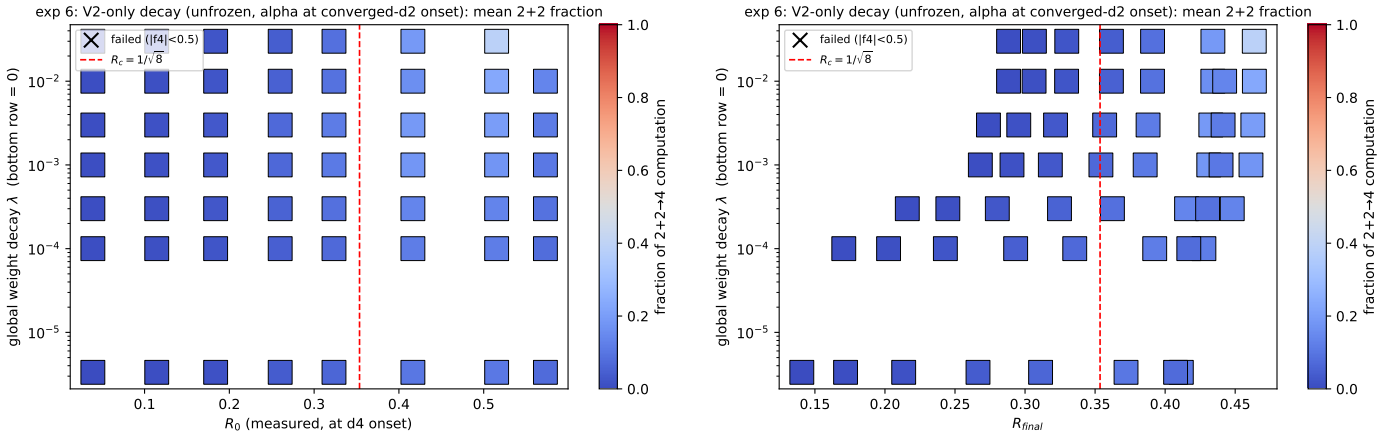


Figure 1: Phase diagrams vs. R_0 (left) and R_{final} (right), colors on the same scale as exps. 2/4. Right of R_c the color now deepens toward composition as λ grows (up the page); missing squares in the top rows are the block-2–bypassed (NaN) cells.

Bottom line. Exp. 4's conclusion sharpens: unfrozen dynamics per se do not kill the compositional phase — *decay on the representation* does. Protecting V_1 restores weight decay's frozen-case role qualitatively (sharpening toward the efficient circuit, pinned at R_c, now with the representation itself drifting upward in accessibility). The remaining factor-$\sim$2 gap to the frozen benchmark is purely dynamical — the direct circuit forms first and is only partially reworked — so closing it should require slowing $d4$ learning (smaller readout lr) or annealed decay, not more decay.

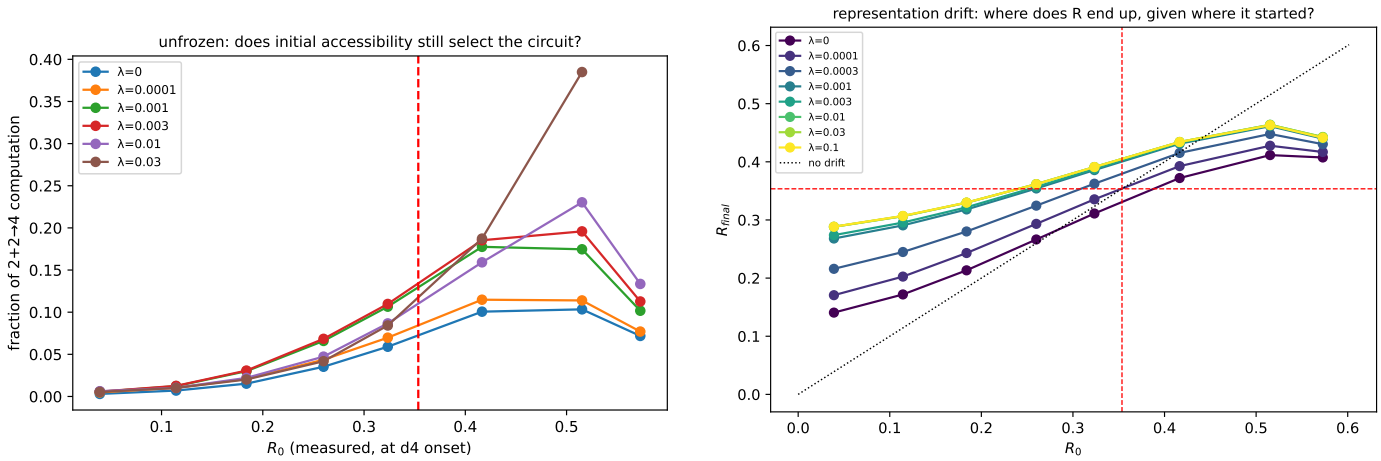


Figure 2: Left: f_{2+2} vs. R_0 at fixed λ — ordered by λ right of R_c (compare exp. 4, where the order was reversed and collapsed to 0). Right: drift map — λ raises R_{final} everywhere; supercritical stays supercritical.

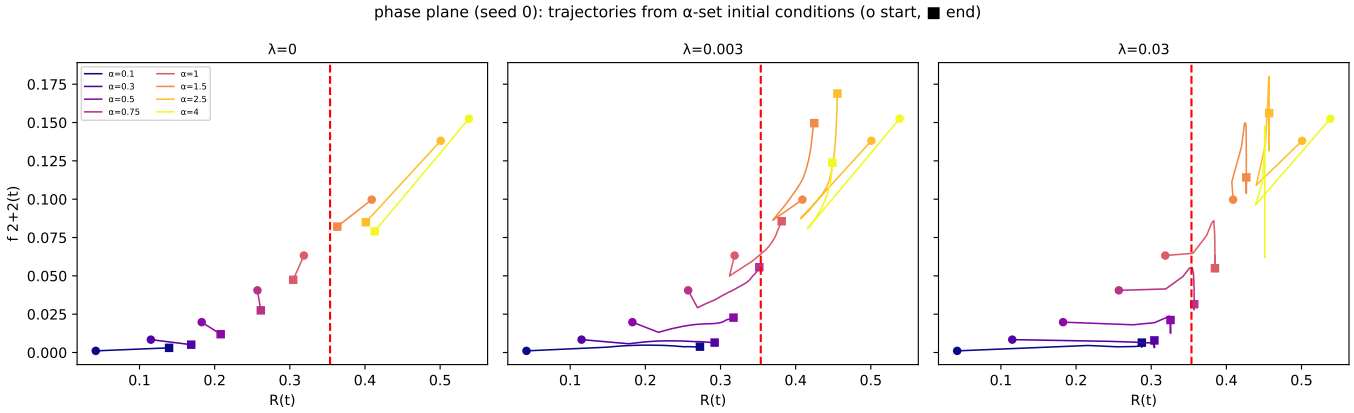


Figure 3: Phase-plane trajectories (seed 0; ∘ start, ■ end). Unlike exp. 4 there are no dismantling loops: under V_2-decay, f_{2+2} grows along the trajectory and $R(t)$ never collapses.

Artifacts: `gdrive:parity/analysis_output/accessibility_alpha_v2wd/` (`results_merged.csv`, 12 worker slices, trajectories, figures, this report); checkpoints: `accessibility_ckpts/pretrain10k_seed{0,1,2}.pt`; code: `analysis/accessibility/run_alpha_unfrozen_experiment.py` (`--wd-block2-only`), `merge_alpha_unfrozen.py`, `scripts/launch_v2wd_workers.sh`. Companions: exps. 1–4 reports.